

Evolution of CPU-GPU Architectures: Towards AI-Driven Heterogeneous Computing Systems

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Abstract— The fast-growing popularity of HPC, AI, and data-oriented applications is leading to increased requirements for computational speedup, scalability, and energy efficiency. While conventional CPU-oriented architecture works well for general-purpose computing, the limitation of such an approach is the inability to effectively process parallel workloads and handle memory bottlenecks while working with today's AI and science loads. As a result, GPUs are introduced as highly parallel computing accelerators, which can provide a significant increase in computational throughput in case of massive simulations, machine learning tasks, and data processing applications. Later, Heterogeneous System Architecture (HSA) is developed to improve cooperation between CPUs and GPUs via the introduction of unified memory and shared virtual address spaces in order to minimize communication costs.

In particular, driven by the rapid development of deep learning models and large language models, computing architectures are moving toward AI-oriented heterogeneous systems that unite CPUs, GPUs, NPU, and other AI accelerators into a single computer platform. In this study, we review the progress of CPU-GPU architecture development, focusing on the limitations of traditional CPU-based architectures, the appearance of GPUs as computing accelerators, HSA for heterogeneous computing, and the impact of AI workloads on processor architectures. Besides, some of the recent architectural advances, like high-bandwidth memory, chiplet integration, and AI-native heterogeneous computing, are described, together with possible areas for future research.

I. INTRODUCTION

The growth of Artificial intelligence (AI) and High-Performance Computing (HPC) has increased the demand significantly for the computational performance. Traditional architectures were mainly based on the Central Processing Units (CPUs). But, with time the complexity of machine learning models, scientific simulations and large-scale data processing has increased due to which the CPU-centric architecture has hit its limits.

To deal these challenges, Graphic Processing Units (GPUs) have shifted from graphic rendering devices into powerful parallel computing accelerators. Together integrated the CPUs and GPUs introduced the heterogeneous computing. Following these advancements, early heterogeneous computing was limited by separate memory space and communication overhead caused by frequent data transfer between CPUs and GPUs. To overcome the same, Heterogeneous System Architecture (HSA) was introduced. It was a unified computing framework that enables shared virtual memory, reduced data movement and efficient processing units. In recent developments, the adoption of AI had influenced the processor design. Today's AI workloads require massive parallel computation, high memory bandwidth.

Simultaneously, the computing architecture are evolving towards AI-driven heterogenous systems that combine CPUs, GPUs, Neural Processing Units (NPUs) and dedicated AI accelerators to improve the performance, scalability and energy efficiency. This paper investigates the evolution of commuting architecture. It discusses the development of GPU acceleration, the role of HAS in improving the CPU-GPU collaboration. Lastly, it explores the upcoming trends the are shaping the next generation of HSA.

II. TRADITIONAL CPU ARCHITECTURES

Traditional CPU architectures have established the foundation of modern computing systems. Based on the von Neumann architecture, the systems use CPUs as the main processing unit. They are optimized for general computing using techniques like multi-core processing, instruction level parallelism and cache hierarchies, making efficient execution of sequential and parallel workloads.

As processing demands increased significantly, CPU performance improved through higher clock frequencies and additional processing cores. Although these changes improved the system performance, but they proved insufficient for highly parallel applications such as large-scale data analytics, scientific computing and AI. The major bottleneck was that, memory access latency restricts the processor performance due to the shared communication between memory and the CPU.

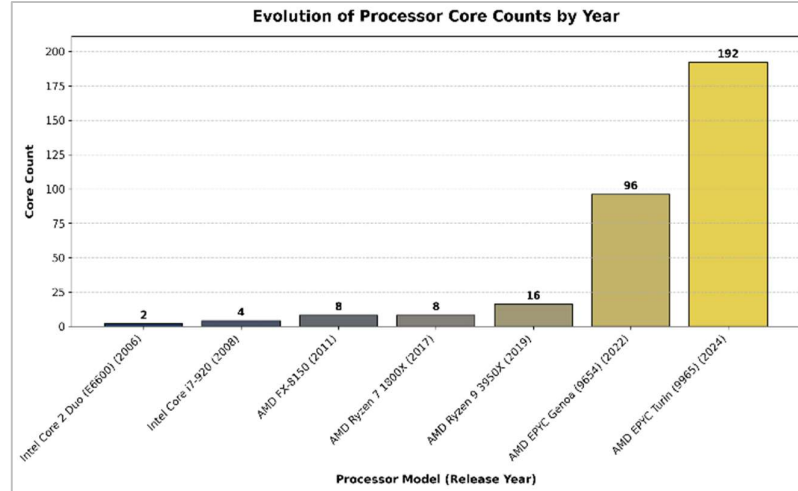


Fig. 1. Evolution of processor core counts in representative CPU architectures (2006–2024)

Moving further, CPUs were mainly designed for low-latency execution rather than massive parallel computation. Therefore, workloads involving large-scale matrix operations and deep learning algorithms cannot utilize CPU resources, resulting in degraded computational efficiency and increased energy consumption.

These overall limitations made the need for specialized parallel processing hardware, leading to the emergence of GPUs. The integration of CPUs and GPUs marked the onset of heterogeneous computing and laid the foundation of AI-driven heterogeneous computing architecture.

III. EVOLUTION OF GPU ARCHITECTURES

The inability of traditional CPU-based systems in dealing with highly parallel workloads was the reason for developing a new class of specialized computing devices – Graphics Processing Units (GPUs). Originally created as computing accelerators for graphics rendering, GPUs later became highly efficient processors that can run thousands of parallel threads at the same time. The massively parallel architecture of these processors made them ideal for a number of compute-intensive tasks like simulation, visualization, data analysis, and even machine learning.

In contrast to the CPUs designed to perform tasks in low-latency and sequential mode, GPUs use thousands of small processors to achieve high throughput. The execution of code in GPUs is based on the SIMT model of execution. Therefore, the same operations can be applied to thousands of data elements simultaneously, and GPUs can significantly outperform CPUs in all types of computations characterized by high level of data-level parallelism.

The unification of CPUs and GPUs represented the emergence of heterogeneous computing, in which each processor performs those operations that suit its architecture the most. CPUs are still responsible for sequential operations, system management, and decision-making, whereas GPUs take care of

parallel computation. Thus, the combination provided significant increase in program performance and allowed current high-performance computing platforms to deal with complex workloads.

However, there were also disadvantages of such systems. They were based on separate memory domains connected via the PCI Express (PCIe) interface. Frequent data transfers from one domain to another led to additional communication costs, increased latency, and decreased system performance. To address these issues, the concept of Heterogeneous System Architecture (HSA) was proposed.

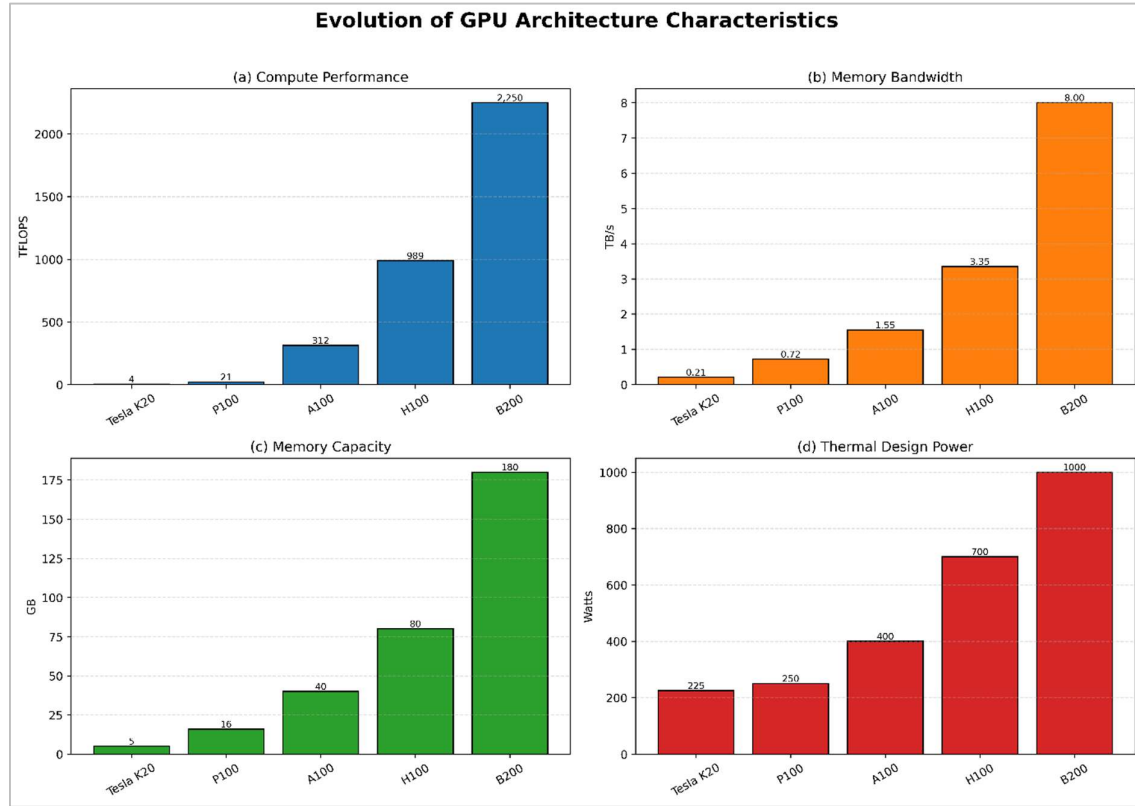


Fig. 2. Evolution of GPU architecture characteristics across representative GPU generations, illustrating improvements in (a) compute performance, (b) memory bandwidth, (c) memory capacity, and (d) thermal design power from Tesla K20 to NVIDIA Blackwell B200

IV. HETEROGENEOUS SYSTEM ARCHITECTURE (HSA)

The use of CPU-GPU heterogeneous computing became much more common, leading to an increase in computational efficiency; however, there were some architectural drawbacks in the earliest heterogeneous systems. The CPUs and GPUs had their own memory areas, and thus, it was necessary to move data through PCI Express (PCIe). These data moves created additional communication overhead and made applications perform less effectively due to higher latency.

In order to solve these problems, the Heterogeneous System Architecture (HSA) was proposed as the standard computing architecture, which makes the CPU-GPU integration much more effective. Instead of considering the GPU as the separate accelerator, it is possible to make the heterogeneous processors work together within a single programming model.

Another characteristic feature of HSA technology is the use of Unified Memory Architecture (UMA) that allows CPUs and GPUs to access the same memory pool without performing any additional data copying operations. Moreover, the architecture is designed based on the use of a shared virtual memory address space, which allows CPUs and GPUs to refer to the same data structures. All of these techniques

help to decrease the overhead related to data transfers and increase the performance of a system due to better coherence of caches.

With the help of these technologies, HSA greatly improves the performance of heterogeneous computing systems when working with scientific computing tasks, machine learning problems, and other data-intensive computing tasks. Despite the fact that modern computing systems went further than HSA was originally proposed, many of the ideas that are at the basis of the architecture still remain important. These ideas play an essential role in the creation of today's computing platforms that combine CPUs, GPUs, and AI accelerators in one environment.

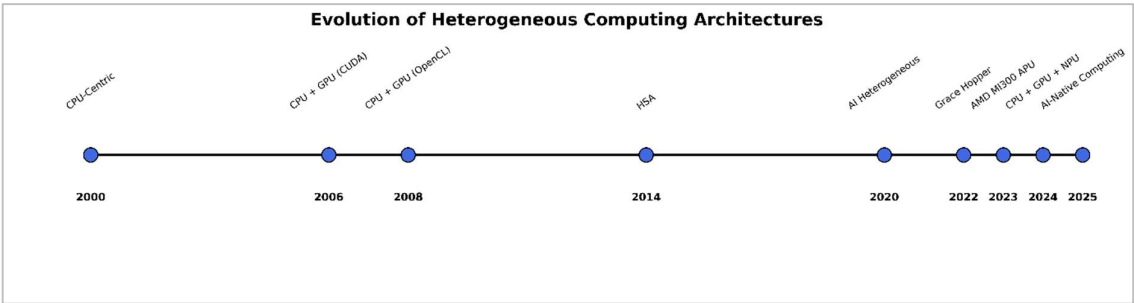


Fig. 3. Evolution of heterogeneous computing architectures from traditional CPU-centric computing to AI-native heterogeneous systems (2000–2025)

V. AI WORKLOADS AND COMPUTATIONAL DEMANDS

With the fast-growing progress of AI technologies, especially of deep learning and generative AI, there have been major changes in computing technologies. Contrary to other scientific and commercial areas where there is a need to perform calculations with relatively small sets of data, AI workloads presuppose processing huge amounts of information and performing billions of mathematical operations. The emergence of transformer networks and LLMs has only boosted the development of AI and, thus, increased the necessity for high-performance computing resources.

The current workloads for AI technologies are mostly based on operations with matrices, tensors, and massive parallel computing. This type of workload requires high computational efficiency, memory bandwidth, and low latency for the effective processing of millions or even billions of parameters. As AI models become more complicated, it becomes impossible to provide enough

parallelism in computation using the traditional CPU-based approach. Even though GPU usage has enhanced the computational speed, it creates additional problems concerning memory capacity, energy consumption, and communication delays.

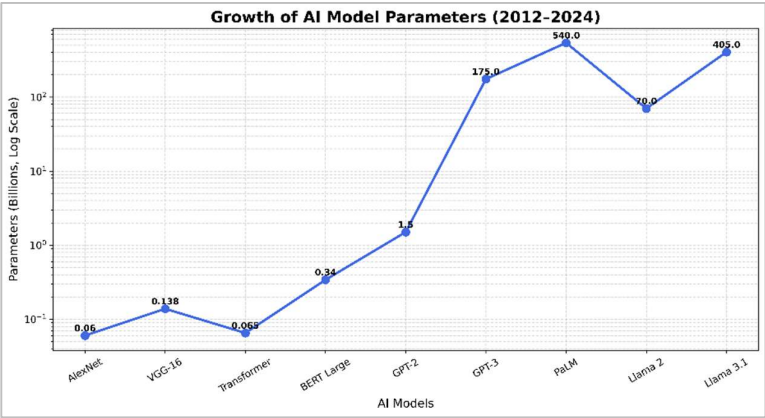


Fig. 4. Evolution of parameter counts in representative AI models (2012–2024) using a logarithmic scale.

Moreover, AI in modern times requires constant transmission of vast amounts of data between the processor and memory, so memory bandwidth becomes an important performance metric. Thus, high bandwidth memory, unified memory architecture, and AI-specific processor modules become key elements of the current computing platform. These approaches decrease the overhead from moving data while increasing computation performance.

The increasing complexity of computations in artificial intelligence has significantly affected the design of processors. In order to meet the current needs, modern computing platforms use not only general-purpose CPU and GPU but also specialized processors like neural processing units and tensor processing units. Thus, the basis of modern heterogeneous computing systems was laid.

VI. AI DRIVEN HETEROGENEOUS COMPUTING ARCHITECTURES

The development of AI technology has played an important role in shaping modern processors' architecture and made heterogeneity become a trend in the evolution of processor design. Although the heterogeneity of Heterogeneous System Architecture (HSA) allowed for more effective interaction between CPU and GPU, the complexity of deep learning and large language models required additional computation capabilities, which resulted in higher memory bandwidth and performance. Modern processor platforms utilize multiple specialized processors tailored specifically for specific computation tasks.

In current processor platforms, there are always at least three different types of processors: CPU, GPU, and NPU. CPUs continue being responsible for control and operating systems, while GPUs handle parallel computations and tensor operations. NPUs perform neural network operations faster and more efficiently than other processors.

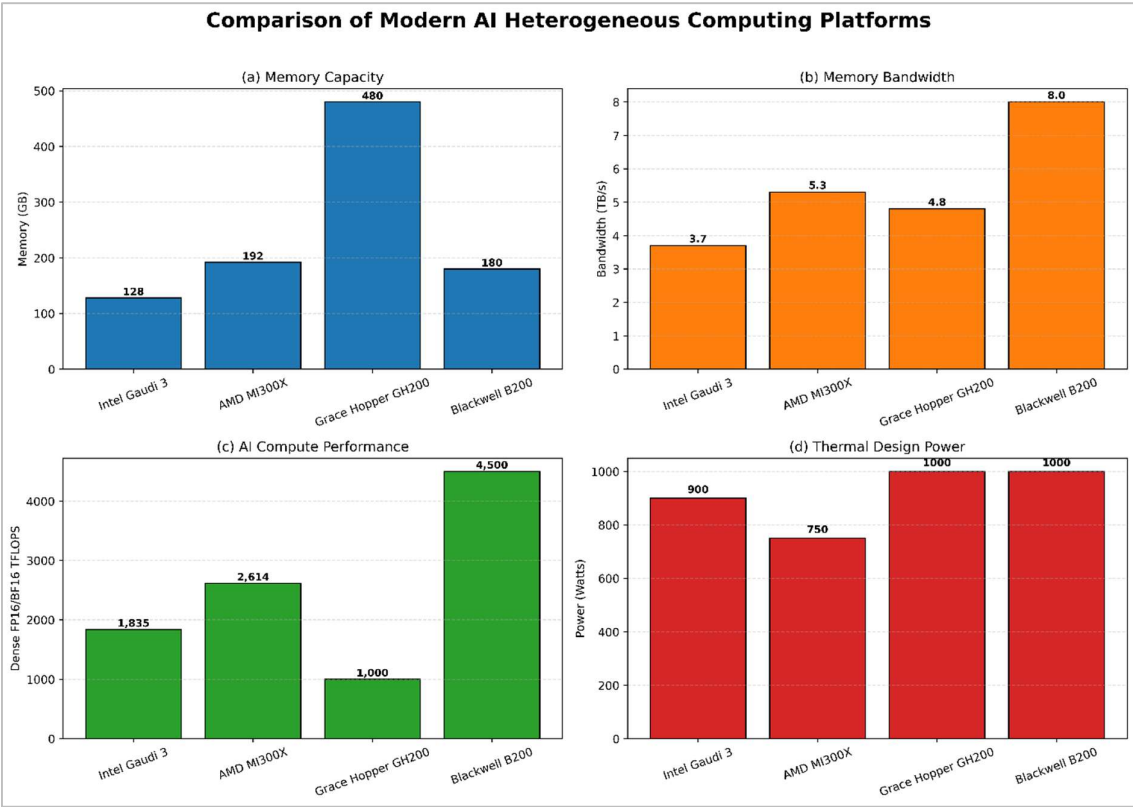


Fig. 5. Comparison of representative AI-driven heterogeneous computing platforms, illustrating (a) memory capacity, (b) memory bandwidth, (c) AI compute performance (Dense FP16/BF16 TFLOPS), and (d) thermal design power. The comparison highlights the architectural advancements that enable modern AI and high-performance computing workloads.

Contemporary architectures are based on such innovations as UMA and HBM that ensure the efficient operation of more sophisticated AI models. The use of unified memory ensures fewer unnecessary data exchanges since processors can operate within the shared memory space. HBM increases memory bandwidth greatly for AI workloads. Both innovations solve the issue of communication latency and increase resource utilization among heterogenous processing units.

All leading semiconductor vendors have introduced new platforms with these innovative architectural features. Thus, the MI300-series of AMD is the platform that combines CPU and GPU chiplets with high-bandwidth memory for HPC and AI tasks. Another example is Grace/Hopper Superchip from NVIDIA that includes Arm-based CPU and high-performance GPU connected through high-speed interconnect for accelerating AI training and scientific computing. In its turn, Intel introduced Core Ultra processors with embedded NPUs to boost AI inference while preserving energy efficiency. All these innovations demonstrate the transition towards integration and optimization of heterogeneous systems for AI tasks.

The union of CPU, GPU, NPU and advanced memory technologies became a breakthrough in computing architecture. It becomes possible to distribute workload between different processors according to their computational capabilities and thus gain performance, scalability, and energy efficiency in a heterogeneous system. It is expected that this innovation will be the basis of future high-performance computing platforms for AI tasks.

VII. FUTURE DEVELOPMENETS

Further developments in AI technology are expected to contribute to the ongoing evolution of heterogeneous computing architecture. With increased complexity of the AI algorithms, future computing systems will require higher computational performance, larger memory sizes, and better power efficiency. Hence, manufacturers of processors are moving on to heterogenous solutions consisting of many specialized processing modules in one computing device.

One of the main trends in the development of modern processors is the introduction of chiplets architecture when CPUs, GPUs, AI engines, memory controllers, and other components are created as individual chiplets and assembled into one package.

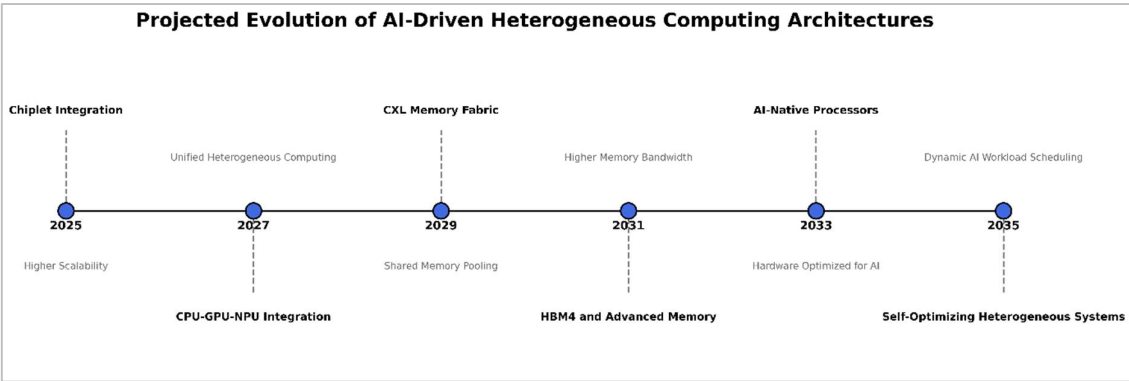


Fig. 6. Projected evolution of AI-driven heterogeneous computing architectures from 2025 to 2035. The roadmap highlights anticipated developments in chiplet integration, unified CPU–GPU–NPU architectures, advanced memory technologies, AI-native processors, and intelligent workload scheduling based on current research and industry trends.

The heterogenous systems of the future will likely place a greater emphasis on memory-centric computing. Techniques like High-Bandwidth Memory (HBM), unified memory architectures, and new interconnects will likely address the issue of latency and improve data transfer between heterogeneous

processors. Such innovations will become increasingly relevant in view of growing amounts of data and models of billions of parameters being processed by AI applications.

The second important trend concerns the creation of processors which will include neural processing units, tensor engines, and special-purpose accelerators as an integral part of computing hardware. Unlike the current generation, where AI is considered a specific workload, the processors of the future will be designed specifically to accommodate the needs of AI, thus ensuring better performance of tasks related to training and inference.

In short, the trend towards heterogeneous computing can be seen in the design of specialized processors, cutting-edge memory technologies, and intelligent task management solutions. Such developments will play an important role in addressing the needs of the next generation of AI applications.

VIII. CONCLUSION

In its development, computing architecture strives to solve the problems of increased requirements to computer capabilities that appear in the field of scientific computations, HPC, and AI. Traditionally, CPU-based architectures laid the groundwork for general-purpose computing architectures but were unable to deliver the required computational throughput due to the inability to provide sufficient parallelism and memory bandwidth in large computational loads. The introduction of GPU enabled the increase in computational throughput by providing parallel processing and Heterogeneous System Architecture (HSA) helped to optimize the use of the resources by means of unified memory and cooperation between CPUs and GPUs.

The fast pace of development of artificial intelligence makes heterogeneous computing architectures based on AI inevitable. This type of computing architecture integrates CPUs, GPUs, NPU, and special AI accelerators into a single computing system that is capable of delivering the performance, memory bandwidth, and energy efficiency needed for advanced AI.

It is expected that the future heterogeneous computing systems will be characterized by the tight coupling of specialized processors, new memory technologies, and intelligent workload scheduling. With time, heterogeneous computing architectures will become an indispensable part of scalable and efficient computing platforms for scientific and industrial needs.

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